Bridgestone SUV tyres - The MMM Case Study

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Contents

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Introduction

- Marketing Mix Modeling (MMM) framework is applied to Bridgestone's SUV tyre portfolio using 104 weeks of synthetic data.
- We define Sales Units as the core KPI and model it as a function of paid media (TV, YouTube, Facebook/Instagram, Search, Display, Influencers, OOH, Radio), owned/1P signals (website sessions, CRM emails), commercial levers (price, promotions, availability), competition, seasonality, and weather.
- The model uses adstock + Hill saturation transforms and Ridge regression for robust performance with an OLS decomposition for contributions.

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Business KPI

- KPI: Weekly Sales Units for Bridgestone SUV replacement tyres.
- Secondary KPI: Sales revenue (units × realized price).

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Key Hypothesis

- H1: Paid media (especially TV and YouTube bursts) drives upper-funnel awareness, translating to lagged and diminishing returns on sales.
- H2: Commercial levers - effective price, promotions/discount depth, and distribution availability are stronger short-term drivers than media.
- H3: Seasonal/monsoon demand and competitor pricing materially shift baseline demand for SUV tyres.

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Data Preparation

- The Synthetic data was prepared with the help of Copilot considering the below aspects.
- Paid Media: TV, YouTube, Facebook/Instagram, Search, Display, Influencers, OOH, Radio (weekly spend).
- Owned / 1P: Website sessions (k), CRM email sends (k).
- Commercial & Market: Price index, Promotions (% discount), Availability (% breadth), Competitor price index, Fuel price index, Rainfall, Seasonality (sin/cos), Trend, Monsoon dummy.

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Feature Engineering

- OLS Assumptions were checked
- Adstock decay per channel

$$a_t = x_t + \lambda a_{t-1}, \text{ channel-specific } \lambda \in [0.3, 0.7].$$

- Hill saturation per channel

$$s(x) = \frac{x^\alpha}{x^\alpha + k^\alpha}, \text{ with } \alpha > 1,\ k \approx \text{60th percentile of adstocked } x.$$

- Seasonality via Fourier (sin/cos) + monsoon dummy
- Price/promotions/availability
- Competition and weather

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Results

- OLS Regression Results before transformation

```
X_raw shape: (104, 20), y length: 104
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.708
Model:                            OLS   Adj. R-squared:                  0.637
Method:                 Least Squares   F-statistic:                     10.05
Date:                Mon, 16 Mar 2026   Prob (F-statistic):           9.06e-15
Time:                        11:16:52   Log-Likelihood:                -618.93
No. Observations:                 104   AIC:                             1280.
Df Residuals:                      83   BIC:                             1335.
Df Model:                          20
Covariance Type:            nonrobust
===========================================================================================
                              coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------------
const                   -1.438e+04   6403.414     -2.246      0.027   -2.71e+04   -1644.751
tv_spend_lakhs             12.2907      2.440      5.037      0.000       7.438      17.144
yt_spend_lakhs             12.6566      3.437      3.682      0.000       5.820      19.493
fb_ig_spend_lakhs           9.2379      3.061      3.018      0.003       3.150      15.325
search_spend_lakhs         27.0537      8.019      3.374      0.001      11.104      43.004
display_spend_lakhs         5.6119      8.925      0.629      0.531     -12.140      23.363
influencer_spend_lakhs     16.4805      7.353      2.241      0.028       1.856      31.105
ooh_spend_lakhs             1.5165      5.277      0.287      0.775      -8.979      12.012
radio_spend_lakhs          14.0052      8.334      1.680      0.097      -2.571      30.582
website_sessions_k          0.0450      0.756      0.060      0.953      -1.458       1.548
...
```

- OLS Regression Results after transformation

```
X_raw shape: (104, 20), y length: 104
                            OLS Regression Results
==============================================================================
Dep. Variable:                      y   R-squared:                       0.980
Model:                            OLS   Adj. R-squared:                  0.975
Method:                 Least Squares   F-statistic:                     205.6
Date:                Mon, 16 Mar 2026   Prob (F-statistic):           4.64e-62
Time:                        11:16:53   Log-Likelihood:                -478.92
No. Observations:                 104   AIC:                             999.8
Df Residuals:                      83   BIC:                             1055.
Df Model:                          20
Covariance Type:            nonrobust
===========================================================================================
                              coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------------
const                     205.7584   1783.060      0.115      0.908   -3340.676    3752.192
tv_m                      510.8091     25.522     20.014      0.000     460.046     561.572
yt_m                      364.2014     22.420     16.245      0.000     319.609     408.794
fb_m                      268.2768     19.270     13.922      0.000     229.949     306.605
search_m                  297.5984     29.893      9.955      0.000     238.142     357.055
display_m                 148.5543     21.800      6.814      0.000     105.195     191.914
influencer_m              179.7985     18.677      9.627      0.000     142.650     216.947
ooh_m                     227.2389     25.482      8.918      0.000     176.557     277.921
radio_m                   143.4799     29.286      4.899      0.000      85.231     201.729
price_index               590.6025    646.168      0.914      0.363    -694.600    1875.805
...
```

Results

- Ridge Regression Results before transformation → 'R2_train': 0.628, 'R2_test': 0.44, 'MAPE_test': 0.032, 'MSE_test': 6123.565
- Ridge Regression Results after transformation → 'R2_train': 0.72, 'R2_test': 0.539, 'MAPE_test': 0.031, 'MSE_test': 5032.305

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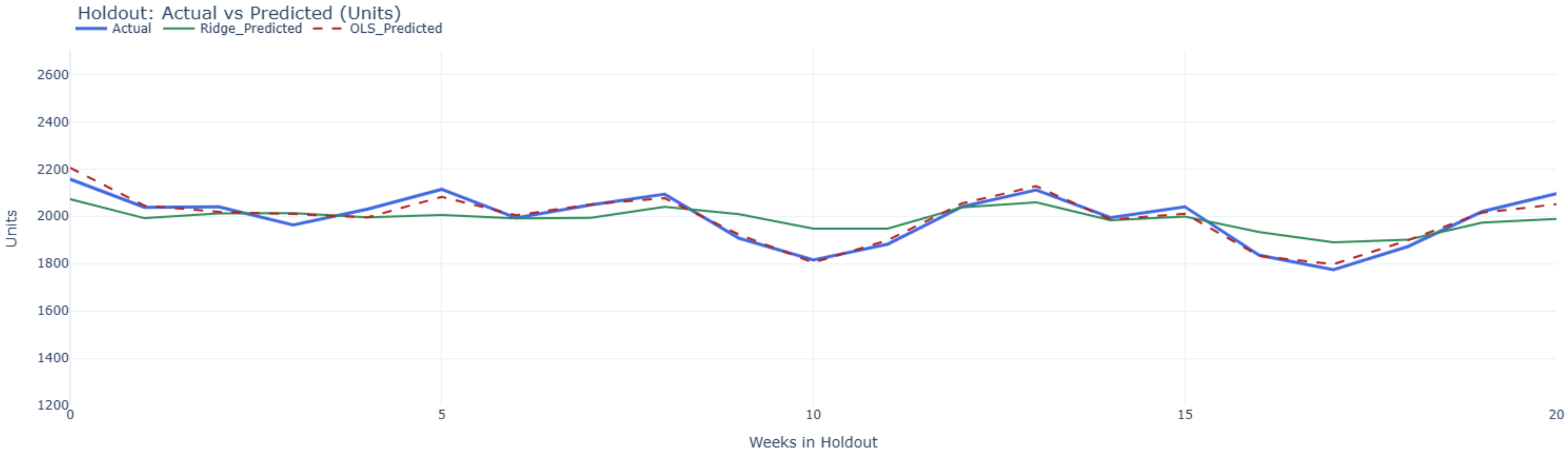
Results

- Why was the Ridge Regression selected over the OLS/Linear Regression?

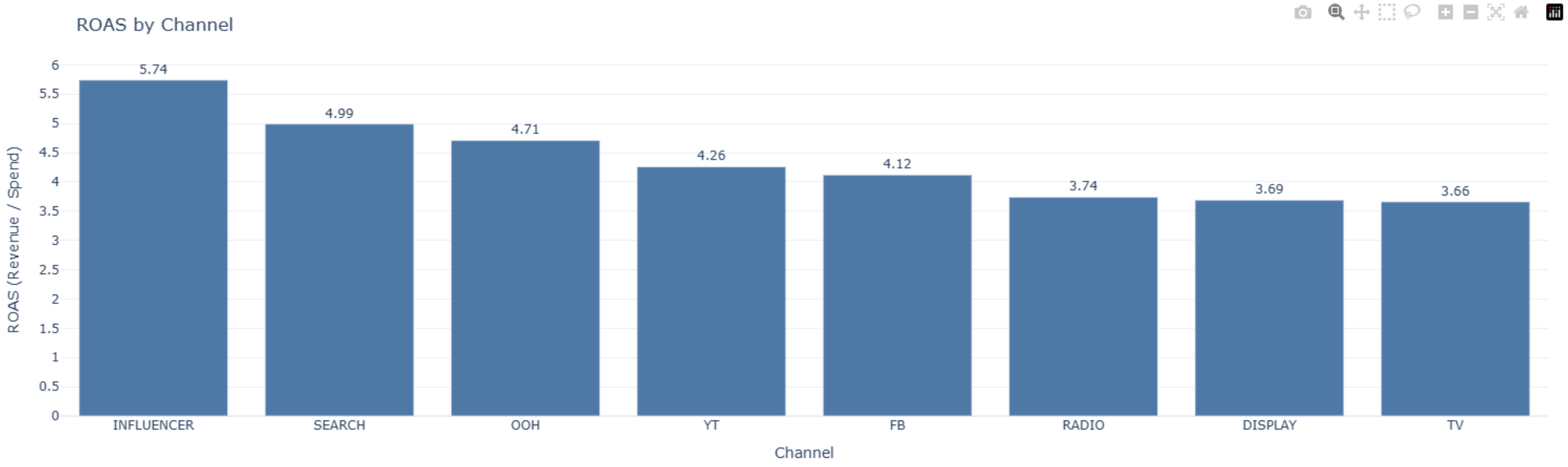
As seen in the previous slides, the results of OLS were not promising, and was overfitting even after the transformation. Hence, to take of the overfitting issue, we used Ridge regression. Refer to the below chart.



Results

- Channel wise ROAS (Revenue/Spend)

Channel	Total Spend (INR)	Attributed Units	Attributed Revenue (INR)	ROAS (Revenue/Spend)
TV	62409048.25	25329.5	2.286271e+08	3.66
YT	37012010.77	17480.5	1.576779e+08	4.26
SEARCH	26740709.28	14790.6	1.334642e+08	4.99
FB	28312761.84	12927.4	1.166440e+08	4.12
OOH	22009421.82	11488.4	1.036621e+08	4.71
INFLUENCER	13855632.40	8803.8	7.951009e+07	5.74
DISPLAY	17025742.40	6962.8	6.280970e+07	3.69
RADIO	16462472.48	6823.6	6.161853e+07	3.74



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Recommendations

- Shift 10–15% from Display/Radio (lower ROAS) and up to 5% from TV (off-pulse) into Search, YouTube, OOH, Influencer.
- Increase promotions in non-monsoon shoulder weeks where price sensitivity is higher.
- Sustain always-on Search; pulse TV/YouTube around monsoon and festive peaks.
- Invest in distribution breadth (availability) which shows strong baseline lift.
- Re-evaluate after 8–10 weeks.

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Thank you!

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